Assignment - 6

Control statement program and looping statement

1. WAP to check whether given number is even/odd/zero.

```
echo "Enter a number:"
read num

if [ $num -eq 0 ]
then echo "Number is ZERO"

elif [ $(($num % 2)) -eq 0 ]
then echo "Number is EVEN"

else echo "Number is ODD"
fi
```

OUTPUT

```
Enter a number : 0
Number is ZERO

Enter a number : 4
Number is EVEN

Enter a number : 5
Number is ODD
```

2.A software company having project team of 3 members names SHIVAM, KIRAN, MIRA.
Input their job experience (in no of years) find out which person who is having highest experience.

```
echo "Enter experience of SHIVAM:"
read S
```

```
echo "Enter experience of KIRAN:"
read K

echo "Enter experience of MIRA:"
read M

if [ $S -ge $K ] && [ $S -ge $M ]
then echo "SHIVAM has highest experience"

elif [ $K -ge $S ] && [ $K -ge $M ]
then echo "KIRAN has highest experience"

else
echo "MIRA has highest experience"
fi
```

OUTPUT

```
Enter experience of SHIVAM :
3
Enter experience of KIRAN :
7
Enter experience of MIRA :
5

KIRAN has highest experience
```

---

3. WAP to input a sufficient value and find the real roots of quadratic equation.

```
echo "Enter a,b,c values:"
read a, b, c
d = $((bb - 4a*c))

if [ $d -gt 0 ]
then
echo "Real and Different Roots"
```

```
elif [ $d -eq 0 ]
then
echo "Real and Equal Roots"
else
echo "Imaginary Roots"
fi
```

OUTPUT

```
Enter a, b, c values:
1 5 6

Real and Different Roots
```

---

4. WAP input a digit within 0 to 6. Display week day.
   Example: 0 for Sunday, 1 for Monday etc. use else if ladder

```
echo "Enter number (0-6):"
read n

if [ $n -eq 0 ]
then echo "Sunday"
elif [ $n -eq 1 ]
then echo "Monday"
elif [ $n -eq 2 ]
then echo "Tuesday"
elif [ $n -eq 3 ]
then echo "Wednesday"
elif [ $n -eq 4 ]
then echo "Thursday"
elif [ $n -eq 5 ]
then echo "Friday"
elif [ $n -eq 6 ]
then echo "Saturday"
else
```

```
echo "Invalid Input"
fi
```

OUTPUT

```
Enter number (0 - 6):
2
Tuesday
```

Loop Control Statement :

i) Loop control is an iterative statement that allows to execute one or a block of statement repeatedly.

ii) logically the loop are of 2 types :

(a) Entry control

(b) Exit control

iii) UNIX shell scripting supports various kind of loop control statement :

a) For loop

b) C-style for loop

c) List-for loop

d) Range-for loop

e) While loop

f) Until loop

g) Select

For loop: For loop is an entry control loop.

Shell script supports 3 different loop like list form, range & C-style form.

List form:

Syntax:

```
for variable in list
do
process the list
done
```

eg :

```
for i in 1 2 3 4 5
do
echo "i = $i"
done
```

Range-loop:

The range is a kind of entry control loop which takes a lower bound and upper bound and it iterates the value.

Syntax :

```
for i in range
do
process the range
done
```

e.g.

```
For i in {1...10}
do
echo "i = $i"
done
```

C-style form:

Syntax :

```
for ((i=0; i<10; i+=2))
do
echo "i = $i"
done
```